

Frame It as Cases: Work That Speaks for Itself

FlyRank AI Internship – AI Fluency

Week 2 | Assignment Code: FL-04

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Track: Machine Learning

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Voice Card

Professional • Clear • Practical • Honest • Evidence-Based • Simple • No Buzzwords

These words define how I want my portfolio and case studies to sound. My goal is to communicate technical work in a way that is clear, authentic, and focused on evidence rather than exaggerated claims.

Case Study 01

FlyRank Machine Learning Internship – Week 1

Problem

The objective of this assignment was to understand how Machine Learning can support real-world SEO decision-making. Using an anonymized search performance dataset provided by FlyRank, I explored how Machine Learning models can identify webpages that are declining in search performance and should be prioritized for content refresh.

The dataset contained approximately **30,000 content pages** with features including search volume, impressions, click-through rate (CTR), average search position, content age, word count, and trend direction.

Rather than simply training a model, the assignment focused on understanding the complete Machine Learning workflow—from data preparation to evaluation and interpretation.

What I Did

I executed the complete starter Machine Learning pipeline in Google Colab and documented each stage of the workflow.

My work included:

- Running the end-to-end Machine Learning pipeline on the provided dataset.
- Comparing a transparent hand-written ranking rule with multiple Machine Learning models, including Logistic Regression, Decision Tree, and Random Forest.
- Analyzing evaluation metrics such as **Precision@50** to understand how effectively each method ranked pages requiring attention.
- Studying Decision Trees to understand how models make predictions using interpretable rules.
- Learning why **client-holdout validation** provides a more realistic evaluation than random train-test splits.
- Investigating **data leakage** by observing how including `trend_pct` dramatically increased performance because it directly revealed the target label.
- Recording observations, organizing notebooks, and maintaining a structured GitHub repository containing executed notebooks, reports, and supporting documentation.

Throughout the assignment, I focused on understanding *why* each stage of the workflow mattered rather than simply executing the provided code.

Challenges

The first challenge was becoming familiar with the development workflow. Since this was my first FlyRank assignment, I needed to understand how to create a repository from the starter template, execute notebooks in Google Colab, synchronize work with GitHub, and submit assignments correctly.

Another challenge was interpreting Machine Learning evaluation rather than simply reading the output. Learning why **Precision@50** was more meaningful than overall accuracy helped me understand that the goal was to rank the highest-priority pages instead of maximizing general classification performance.

The most valuable learning experience came from studying **data leakage**. Initially, the nearly perfect performance achieved by including `trend_pct` appeared impressive. After analyzing the notebook, I realized that the feature effectively revealed the target label, making the evaluation unrealistic. This reinforced the importance of proper feature selection and honest model validation.

Outcome

The assignment was completed successfully with all required notebooks executed and documented.

Key outcomes included:

- Successfully executed a complete Machine Learning pipeline using real anonymized search data.
- Compared baseline rules with Machine Learning models using meaningful evaluation metrics.
- Developed a stronger understanding of feature engineering, validation strategies, and model evaluation.
- Learned to recognize and avoid data leakage.
- Improved practical skills in Python, Google Colab, GitHub, Pandas, NumPy, and Scikit-learn.
- Built a reproducible GitHub portfolio documenting each stage of the project.

More importantly, the assignment shifted my perspective from simply building models to understanding how Machine Learning supports real-world business decisions.

About Me

I am a Software Engineering student with a strong interest in Artificial Intelligence, Machine Learning, and Data Science. I enjoy solving practical problems through data-driven solutions while documenting my work in a professional and reproducible manner. My goal is to continuously improve my technical skills through hands-on projects and build a portfolio that demonstrates real-world problem-solving ability.

Contact / Call to Action

Thank you for reviewing this case study.

If you would like to discuss Machine Learning, Artificial Intelligence, or potential internship opportunities, feel free to connect with me through my GitHub portfolio or LinkedIn profile.

Before & After

Generic AI Version

I completed a Machine Learning project using Python and built models to analyze data.

My Edited Version

I executed a complete Machine Learning workflow using an anonymized dataset of approximately 30,000 webpages to understand how predictive models can support SEO content decisions. I compared rule-based and machine learning approaches, analyzed evaluation metrics such as Precision@50, investigated data leakage, interpreted Decision Trees, and documented the entire workflow in a reproducible GitHub repository. Rather than focusing only on model performance, I developed a deeper understanding of responsible model evaluation and practical Machine Learning applications.

Reflection

This assignment demonstrated that effective Machine Learning is not only about selecting algorithms—it is about asking the right questions, choosing reliable features, evaluating models honestly, and communicating results clearly. Building reproducible workflows and understanding model behavior are skills that will continue to guide my future projects throughout the FlyRank AI Internship and beyond.